Introduction to MIMIC

- Medical Information Mart for Intensive Care (MIMIC)
- Large, open-access ICU database
- Developed by MIT & Beth Israel Deaconess Medical Center
- Used worldwide for clinical & ML research

MIMIC Versions

- MIMIC-III: Older, widely cited
- MIMIC-IV: Updated schema, cleaner structure
- Contains de-identified real ICU data

What Data Does MIMIC Contain?

- Patient demographics
- Hospital admissions
- ICU stays
- Vitals, labs, medications
- Clinical events & procedures
- Diagnosis(ICD Coding)

Core Identifiers in MIMIC

- subject_id → Patient
- hadm_id → Hospital admission
- stay_id → ICU stay
- One patient can have multiple admissions and ICU stays

Abstract

MIMIC-III ('Medical Information Mart for Intensive Care') is a large, single-center database comprising information relating to patients admitted to critical care units at a large tertiary care hospital. Data includes vital signs, medications, laboratory measurements, observations and notes charted by care providers, fluid balance, procedure codes, diagnostic codes, imaging reports, hospital length of stay, survival data, and more. The database supports applications including academic and industrial research, quality improvement initiatives, and higher education coursework.

Design Type(s)	data integration objective
Measurement Type(s)	Demographics • clinical measurement • intervention • Billing • Medical History Dictionary • Pharmacotherapy • clinical laboratory test • medical data
Technology Type(s)	Electronic Medical Record • Medical Record • Electronic Billing System • Medical Coding Process Document • Free Text Format
Factor Type(s)	
Sample Characteristic(s)	Homo sapiens

Table 1 Details of the MIMIC-III patient population by first critical care unit on hospital admission for patients aged 16 years and above.

From: MIMIC-III, a freely accessible critical care database

Critical care unit	CCU	CSRU	MICU	SICU	TSICU	Total
Distinct patients, no. (% of total admissions)	5,674 (14.7%)	8,091 (20.9%)	13,649 (35.4%)	6,372 (16.5%)	4,811 (12.5%)	38,597 (100%)
Hospital admissions, no. (% of total admissions)	7,258 (14.6%)	9,156 (18.4%)	19,770 (39.7%)	8,110 (16.3%)	5,491 (11.0%)	49,785 (100%)
Distinct ICU stays, no. (% of total admissions)	7,726 (14.5%)	9,854 (18.4%)	21,087 (39.5%)	8,891 (16.6%)	5,865 (11.0%)	53,423 (100%)
Age, years, median (Q1-Q3)	70.1 (58.4–80.5)	67.6 (57.6–76.7)	64.9 (51.7–78.2)	63.6 (51.4–76.5)	59.9 (42.9–75.7)	65.8 (52.8–77.8)
Gender, male, % of unit stays	4,203 (57.9%)	6,000 (65.5%)	10,193 (51.6%)	4,251 (52.4%)	3,336 (60.7%)	27,983 (55.9%)
ICU length of stay, median days (Q1-Q3)	2.2 (1.2–4.1)	2.2 (1.2–4.0)	2.1 (1.2–4.1)	2.3 (1.3–4.9)	2.1 (1.2–4.6)	2.1 (1.2–4.6)
Hospital length of stay, median days (Q1-Q3)	5.8 (3.1–10.0)	7.4 (5.2–11.4)	6.4 (3.7–11.7)	7.9 (4.4–14.2)	7.4 (4.1–13.6)	6.9 (4.1–11.9)
ICU mortality, percent of unit stays	685 (8.9%)	353 (3.6%)	2,222 (10.5%)	813 (9.1%)	492 (8.4%)	4,565 (8.5%)
Hospital mortality, percent of unit stays	817 (11.3%)	424 (4.6%)	2,859 (14.5%)	1,020 (12.6%)	628 (11.4%)	5,748 (11.5%)

CCU is Coronary Care Unit; CSRU is Cardiac Surgery Recovery Unit; MICU is Medical Intensive Care Unit; SICU is Surgical Intensive Care Unit; TSICU is Trauma Surgical Intensive Care Unit.

Table 2 Distribution of primary International Classification of Diseases, 9th Edition (ICD-9) codes by care unit for patients aged 16 years and above.

From: MIMIC-III, a freely accessible critical care database

Critical care unit	CCU stays, No. (% by unit)	CSRU stays, No. (% by unit)	MICU stays, No. (% by unit)	SICU stays, No. (% by unit)	TSICU stays, No. (% by unit)	Total stays, No. (% by unit)
Infectious and parasitic diseases, i.e., septicemia, other infectious and parasitic diseases, etc., (001–139)	305 (4.2%)	72 (0.8%)	3,229 (16.7%)	448 (5.6%)	152 (2.8%)	4,206 (8.6%)
Neoplasms of digestive organs and intrathoracic organs, etc., (140–239)	126 (1.8%)	287 (3.2%)	1,415 (7.3%)	1,225 (15.3%)	466 (8.6%)	3,519 (7.2%)
Endocrine, nutritional, metabolic, and immunity (240–279)	104 (1.4%)	36 (0.4%)	985 (5.1%)	178 (2.2%)	54 (1.0%)	1,357 (2.8%)
Diseases of the circulatory system, i.e., ischemic heart diseases, diseases of pulmonary circulation, dysrhythmias, heart failure, cerebrovascular diseases, etc., (390–459)	5,131 (71.4%)	7,138 (78.6%)	2,638 (13.6%)	2,356 (29.5%)	684 (12.6%)	17,947 (36.6%)
Pulmonary diseases, i.e., pneumonia and influenza, chronic obstructive pulmonary disease, etc., (460–519)	416 (5.8%)	141 (1.6%)	3,393 (17.5%)	390 (4.9%)	225 (4.1%)	4,565 (9.3%)
Diseases of the digestive system (520–579)	264 (3.7%)	157 (1.7%)	3,046 (15.7%)	1,193 (14.9%)	440 (8.1%)	5,100 (10.4%)
Diseases of the genitourinary system, i.e., nephritis, nephrotic syndrome, nephrosis, and other diseases of the genitourinary system (580–629)	130 (1.8%)	14 (0.2%)	738 (3.8%)	101 (1.3%)	31 (0.6%)	1,014 (2.1%)
Trauma (800–959)	97 (1.3%)	494 (5.4%)	480 (2.5%)	836 (10.5%)	2,809 (51.7%)	4,716 (9.6%)
Poisoning by drugs and biological substances (960–979)	50 (0.7%)	2 (0.0%)	584 (3.0%)	58 (0.7%)	11 (0.2%)	705 (1.4%)
Other	565 (7.9%)	739 (8.1%)	2,883 (14.9%)	1,204 (15.1%)	563 (10.4%)	5,954 (12.1%)
Total	7,188 (14.6%)	9,080 (18.5%)	19,391 (39.5%)	7,989 (16.3%)	5,435 (11.1%)	49,083 (100%)

Table 3 Classes of data available in the MIMIC-III critical care database.

From: MIMIC-III, a freely accessible critical care database

Class of data	Description
Billing	Coded data recorded primarily for billing and administrative purposes. Includes Current Procedural Terminology (CPT) codes, Diagnosis-Related Group (DRG) codes, and International Classification of Diseases (ICD) codes.
Descriptive	Demographic detail, admission and discharge times, and dates of death.
Dictionary	Look-up tables for cross referencing concept identifiers (for example, International Classification of Diseases (ICD) codes) with associated labels.
Interventions	Procedures such as dialysis, imaging studies, and placement of lines.
Laboratory	Blood chemistry, hematology, urine analysis, and microbiology test results.
Medications	Administration records of intravenous medications and medication orders.
Notes	Free text notes such as provider progress notes and hospital discharge summaries.
Physiologic	Nurse-verified vital signs, approximately hourly (e.g., heart rate, blood pressure, respiratory rate).
Reports	Free text reports of electrocardiogram and imaging studies.

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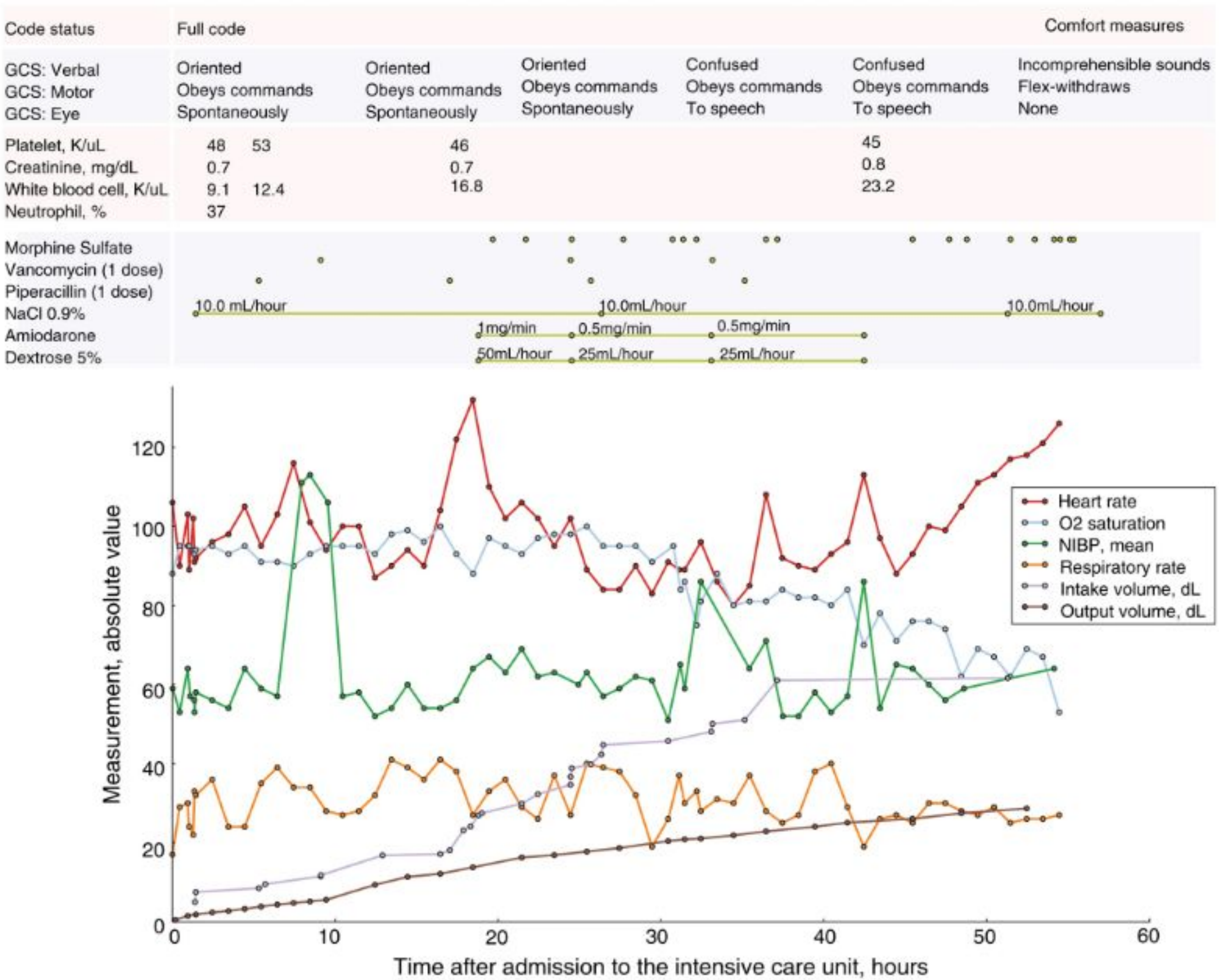
Figure 2: Sample data for a single patient stay in a medical intensive care unit.

From: MIMIC-III, a freely accessible critical care database



Deidentification

Before data was incorporated into the MIMIC-III database, it was first deidentified in accordance with Health Insurance Portability and Accountability Act (HIPAA) standards using structured data cleansing and date shifting. The deidentification process for structured data required the removal of all eighteen of the identifying data elements listed in HIPAA, including fields such as patient name, telephone number, address, and dates. In particular, dates were shifted into the future by a random offset for each individual patient in a consistent manner to preserve intervals, resulting in stays which occur sometime between the years 2100 and 2200. Time of day, day of the week, and approximate seasonality were conserved during date shifting. Dates of birth for patients aged over 89 were shifted to obscure their true age and comply with HIPAA regulations: these patients appear in the database with ages of over 300 years.

Protected health information was removed from free text fields, such as diagnostic reports and physician notes, using a rigorously evaluated deidentification system based on extensive dictionary look-ups and pattern-matching with regular expressions[6]. The components of this deidentification system are continually expanded as new data is acquired.

Figure 1: Overview of the MIMIC-III critical care database.

From: MIMIC-III, a freely accessible critical care database

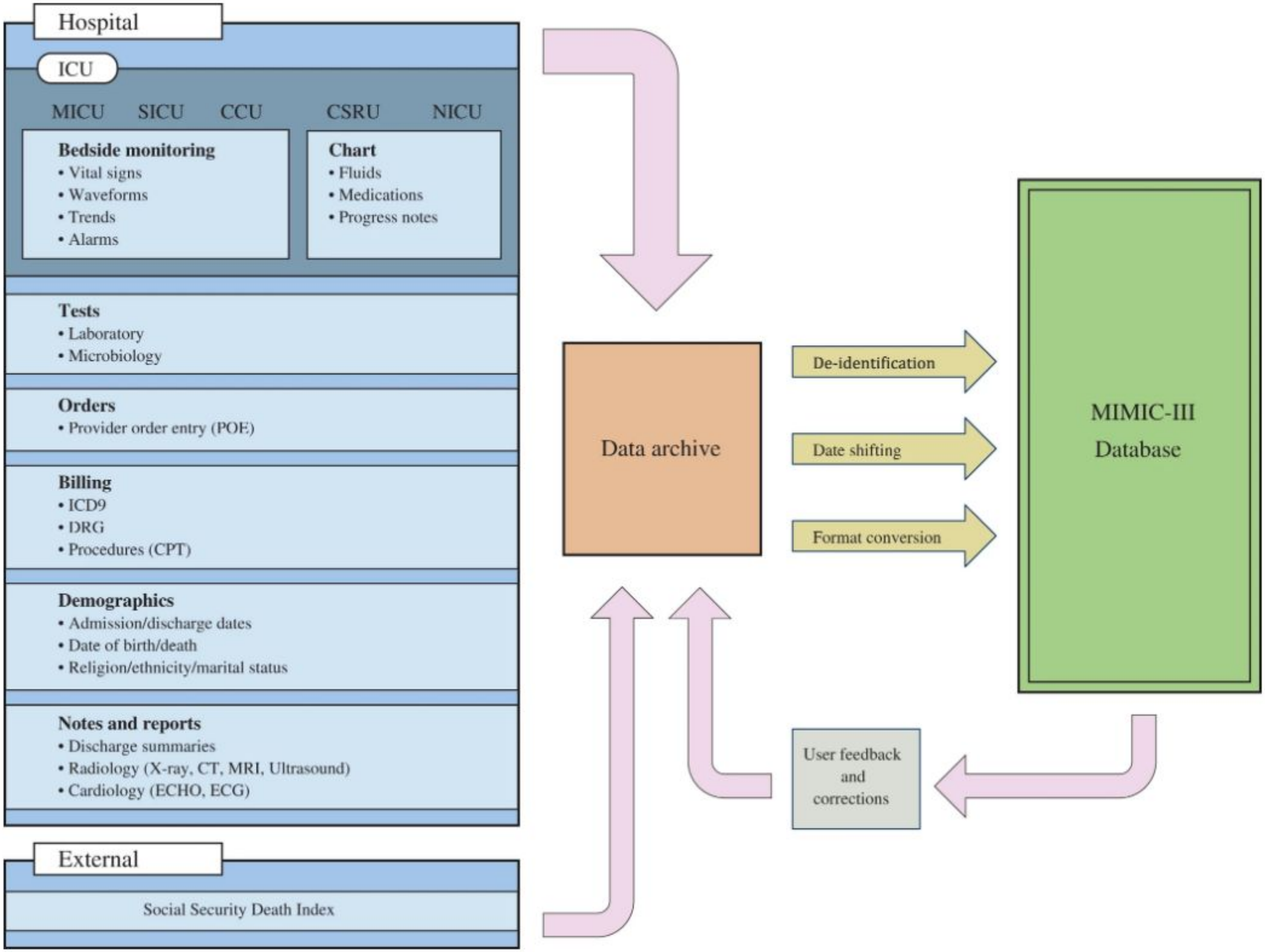
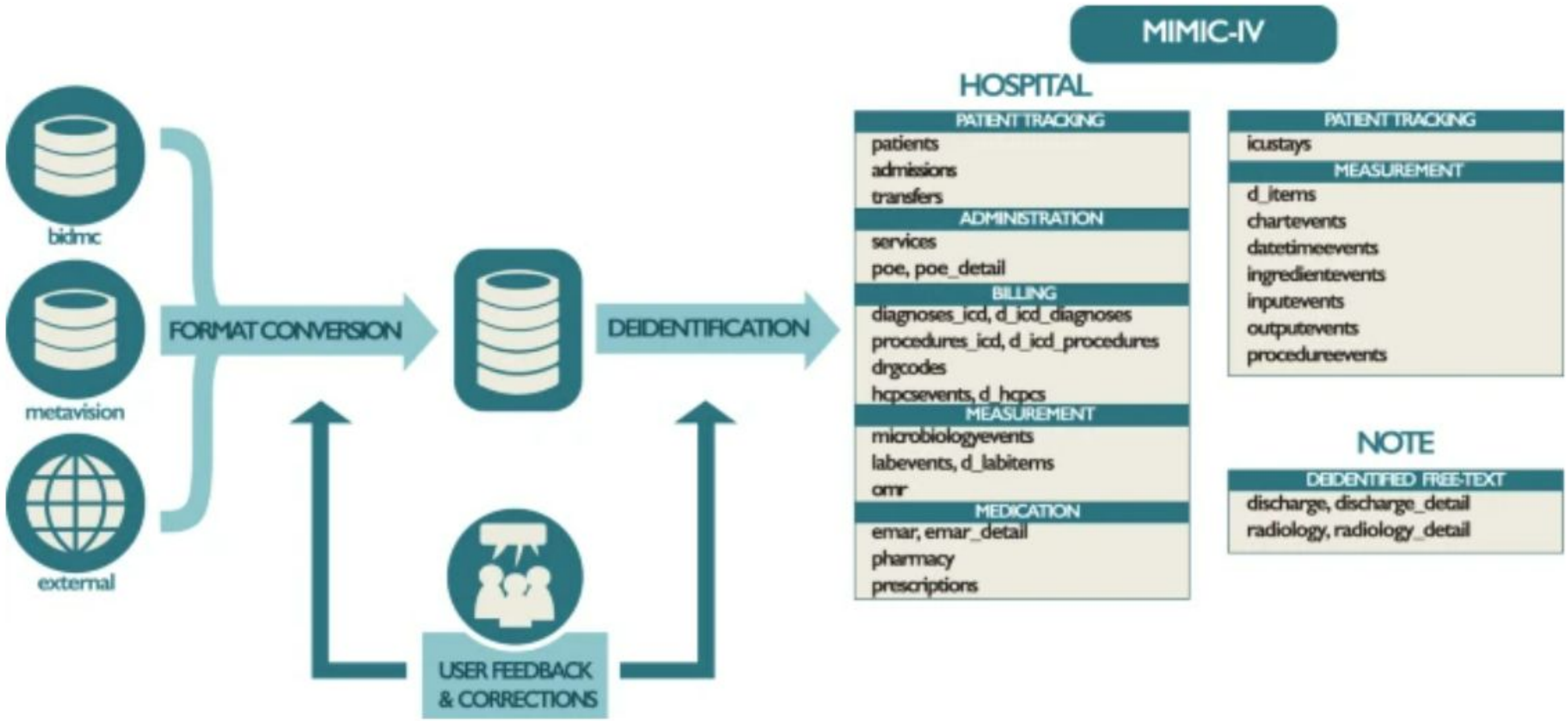


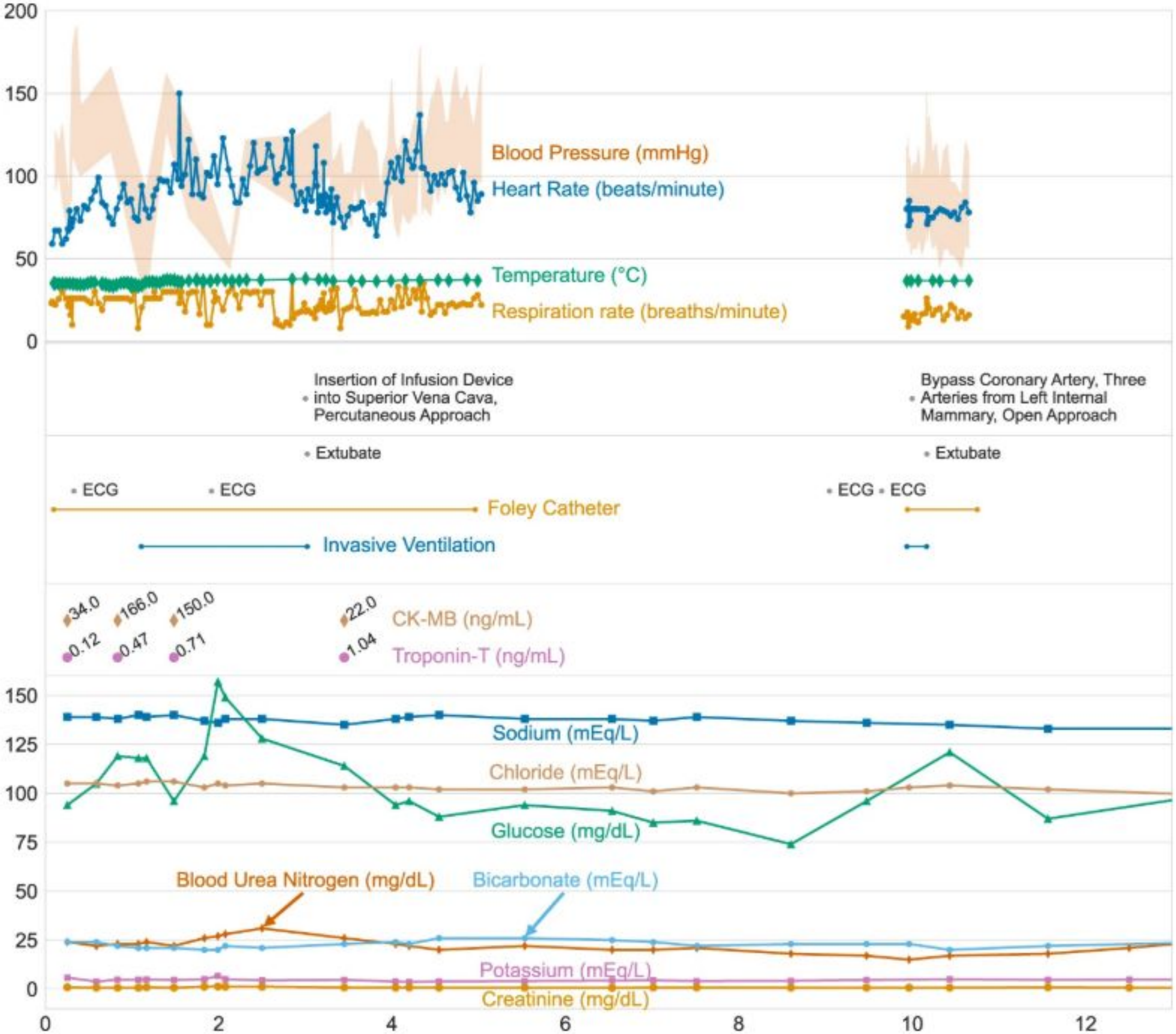
Fig. 1



MIMIC-IV
HOSPITAL
PATIENT TRACKING
patients
admissions
transfers
ADMINISTRATION
services
poe, poe_detail
BILLING
diagnoses_icd, d_icd_diagnoses
procedures_icd, d_icd_procedures
drgcodes
hcpcsevents, d_hcpcs
MEASUREMENT
microbiologyevents
labevents, d_labitems
omr
MEDICATION
emar, emar_detail
pharmacy
prescriptions
PATIENT TRACKING
icustays
MEASUREMENT
d_items
chartevents
datetimeevents
ingredientevents
inputevents
outputevents
procedureevents
NOTE
DEIDENTIFIED FREE-TEXT
discharge, discharge_detail
radiology, radiology_detail
bidmc
metavision
external
FORMAT CONVERSION
DEIDENTIFICATION
USER FEEDBACK
& CORRECTIONS

Fig. 3

From: MIMIC-IV, a freely accessible electronic health record dataset



Visualization of data within MIMIC-IV for a single patient's hospitalization: `hadm_id 28503629`. Three vertically stacked panels highlight the variety of information available. Vital signs are shown in the top panel: note the frequency of data collection for temperature is much higher at the start of the ICU stay due to the use of targeted temperature management. Procedures from multiple sources are shown in the middle panel, including from billing information, the provider order entry system, as well as the ICU information system. The bottom panel displays patient laboratory measurements. Note that while frequent vital signs are only available when the patient is in the ICU, laboratory measures are available throughout their hospitalization.

Main MIMIC-IV Modules

- patients
- hosp (admissions, diagnoses)
- icu (chartevents, icustays)
- labevents
- emar / prescriptions

Data Leakage

- Data leakage occurs when information from the test set influences the training set
- Clinical patterns are highly Patient specific.
- Random row level split can leak information
- Always split by patient not by rows.

Ethics & Access

- Requires CITI training
- Data is de-identified
- Must be used responsibly

Ethics & Access

- Medical data represents human lives.
- Researchers hold moral responsibility.
- Deidentification(**HIPPA Standards**)reduces risk but do not eliminate it.
- **“Medical data are not numbers — they are traces of vulnerable moments in human lives.”**

Ethical Principles

- Respect for Persons(Privacy, Autonomy, Confidentiality)
- **Non-maleficence**
- **Justice(Fairness)**

Example Research Questions

- Mortality prediction
- Sepsis detection
- Length of stay prediction
- Drug–drug interactions
- Readmission Predictions

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Computers in Biology and Medicine

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Predicting sepsis with a recurrent neural network using the MIMIC III database

Matthieu Scherpf [a], Felix Gräßer [a], Hagen Malberg [a], Sebastian Zaunseder [b]

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https://doi.org/10.1016/j.compbiomed.2019.103395

Highlights

- Sepsis prediction can be improved using recurrent neural networks.

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Detecting Cardiovascular Morbidity using MIMIC-III Database: Assessing Three Data Imbalance Handling Techniques

Authors: Amina Zahar, Zineb Belkho, Omar El Kalkha, Fadoua Khennou, Othmane El Meslouhi | Authors Info & Claims

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Prediction of Intensive Care Unit Length of Stay in the MIMIC-IV Dataset

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ADVANCED SEARCH

ECG Abnormality Detection Using MIMIC-IV-ECG Data Via Supervised Contrastive Learning

Publisher: IEEE

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Abstract:
Electrocardiogram data provide a tremendous opportunity for the detection of various types of cardiac arrhythmia. Recent advancement in ubiquitous wearable devices with incorporated ECG sensors offers an

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Key Takeaway

- MIMIC reflects real ICU complexity
- Powerful dataset when used correctly
- Focus on clinical reasoning, not just models